INCH - POUND

MIL-P-47276B (MI)
25 September 1991
SUPERSEDING
MIL-P-47276A (MI)
23 May 1988

MILITARY SPECIFICATION

PRIMER, BONDING

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of liquid organic nitrilephenolic resin primer, specified herein as the primer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8040

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2.1.1 Specifications, standard, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

MMM-A-132	-	Adhesives, Heat Resistant, Airframe Structural, Metal-to-Metal

MILITARY

MIL-P-116	-	Preservation, Methods of

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1190	-	Minimum Guidelines for Level C preservation, packing and marking
MIL-STD-2073	-	DOD Material, Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 1200	-	Viscosity by Ford Viscosity Cup, Standard Test Method for
ASTM D 1259	-	Nonvolatile Content of Resin Solutions, Standard Test Methods for
ASTM D 1475	-	Density of Paint, Varnish, Lacquer, and Related Products, Standard Test Method for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1137.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Materials. The primer shall consist of a liquid organic nitrile-phenolic resin of such composition that will yield a product conforming to all requirements specified herein.

3.3 Toxicity - flammability. Unless otherwise specified all known toxic and flammable materials in the primer shall be identified on the unit and shipping containers (see 5.2). Precautions to be observed in the use of this primer shall be marked on the containers. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting officer to the appropriate departmental medical service who will act as an advisor to the contracting agency.

3.4 Physical properties.

3.4.1 Solids content. The solids content of the primer shall be 15 percent (%) ± 5% by weight (see 4.7.2).

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3.4.2 Viscosity. The viscosity of the primer shall yield a value not greater than 36 seconds at 27° Celsius (C) (see 4.7.3).

3.4.3 Density. The density of the primer shall be 7.2 ± 0.2 pounds per gallon (see 4.7.4).

3.4.4 Dilution stability. A mixture of the primer with methyl isobutyl ketone shall exhibit no curdling, precipitation or separation (see 4.7.5).

3.4.5 Surface finish. The coated panel surface shall have a smooth, fine-grained finish, free from streaks, blisters, built-up edges, and other surface irregularities (see 4.7.6.2).

3.4.6 Film thickness. A single spraycoat of the primer shall dry to a thickness of 0.0001 to 0.0003 inch (see 4.7.7).

3.4.7 Shear strength. Joints which are primed with primer conforming to this specification and bonded with film adhesive conforming to MMM-A-132, type I, class 2 form F, group 3 (0.208×10^{-3} pounds force per square inch (lbf/in^2)) shall show shear strengths as specified in table I (see 4.7.8).

TABLE I. Shear strength requirements.

Temperature	Number of panels tested	Shear strength, average, minimum	Shear, strength, individual, minimum
24°C	4	2500 lbf/in^2	2200 lbf/in^2
82°C	4	1250 lbf/in^2	---
-55°C	4	2500 lbf/in^2	2200 lbf/in^2

3.5 Workmanship. The components shall be mixed and processed in such a way as to ensure compliance with the requirements of this specification. Freshly-opened containers of the primer shall be free from grit, sediment, and suspended matter (see 4.7.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the

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contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed under conditions of room ambient pressure, temperature, and humidity.

4.4 First article inspection. Unless otherwise specified, three containers of primer shall be randomly selected from the first lot manufactured under each contract or purchase order. A sample from each container shall be subjected to each of the quality conformance inspections specified in table II in the order listed. The first article sample shall be taken from a batch of primer which has been produced by the same processes as proposed for all of the primer furnished under the same contract or purchase order.

4.5 Quality conformance inspection. Quality conformance inspections shall be as specified in table II.

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TABLE II. Quality conformance inspection.

Examination or test	Requirement paragraph	Method paragraph
Solids content	3.4.1	4.7.2
Viscosity	3.4.2	4.7.3
Density	3.4.3	4.7.4
Dilution stability	3.4.4	4.7.4
Surface finish	3.4.5	4.7.6.2
Film thickness	3.4.6	4.7.7
Shear strength	3.4.7	4.7.8
Workmanship	3.5	4.7.1

4.6 Inspection lot. An inspection lot shall consist of a single batch of primer offered for inspection at one time. Each batch shall have been produced in a single vat during a continuous production period not greater than 24 hours with no change in process or raw materials.

4.7 Test methods.

4.7.1 Visual examination for workmanship. The primer in freshly opened containers shall be visually examined to determine compliance with requirements specified in 3.5. If necessary, small quantities of the primer shall be poured onto clean glass surfaces for examination. Once opened for inspection, the primer shall be used within an eight hour period or shall be discarded.

4.7.2 Solids content test. The solids content of the primer shall be measured in accordance with ASTM D 1259 to verify compliance with requirements specified in 3.4.1.

4.7.3 Viscosity test. The viscosity of the primer shall be measured with a Ford Cup Viscosimeter with 0.134 inch orifice, in accordance with ASTM D 1200, to verify compliance with requirements specified in 3.4.2.

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4.7.4 Density test. The density of the primer shall be measured in accordance with ASTM D 1475, to verify compliance with requirements specified in 3.4.3.

4.7.5 Dilution stability. The dilution stability of the primer shall be determined as specified herein. Mix 50 ± 1 milliliters (ml) of the primer with 50 ± 1 ml of methyl isobutyl ketone. Place the mixture in a well-stoppered glass container, and store it for 4 to 4.5 hours at 23 + 2°C. At the end of this period, examine the material for compliance with requirements specified in 3.4.4.

4.7.6 Surface finish tests.

4.7.6.1 Preparation of test panels. Twelve aluminum test panels shall be prepared as specified in MMM-A-132 with the following exceptions:

a. The aluminum shall be wiped clean, de-greased, cleaned for 10 minutes in a solution containing 39 parts water by weight, 10 parts Baume sulphuric acid and one part sodium dichromate, at a temperature of 60 to 71°C, rinsed with tap water and then distilled water, and dried at 51 to 66°C for 15 to 30 minutes in a forced-air circulation oven.

b. The primer shall be applied by spray gun to the area to be bonded, and after air-drying for 10 to 15 minutes, the primed parts shall be placed in a circulating air type oven at 79 ± 15°C for 30 minutes.

c. After the primer has dried, pairs of panels shall be bonded together by applying film adhesive conforming to MMM-A-132 type I, class 2 form F, group 3 (0.208×10^{-3} lbf/in^2) between the primed surfaces and curing the bonds under a pressure of 150 ± 5 lbf/in^2 and a temperature of 160 + 3, -5°C for 180 ± 5 minutes.

4.7.6.2 Visual examination for surface finish. After the panels have been cured, they shall be visually inspected for compliance with the surface finish requirements specified in 3.4.5.

4.7.7 Film thickness test. The thickness of the film on each test panel shall be measured to verify compliance with requirements specified in 3.4.6.

4.7.8 Shear strength test. The test panels used for visual inspection shall be subjected to the shear strength tests specified in MMM-A-132 ten minutes after reaching equilibrium at the required temperature to verify compliance with requirements specified in table I.

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4.7.9 Toxicity certification. The supplier shall, if specified in the contract or order (see 6.2), certify that the primer complies with requirements specified in 3.3. As specified in the contract or order (see 6.2), the primer shall be tested to determine whether non-permitted chemicals are included in the material.

4.8 Inspection of packaging. Except when commercial packaging is specified, the sampling and inspection of the preservation and interior package marking shall be in accordance with groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification shown in section 5. The inspection of marking for shipment and storage shall be in accordance with MIL-STD-129. The inspection of commercial packaging shall be as specified in the contract (see 6.2).

5. PACKAGING

5.1 Level A or B. Preservation, Packaging and Packing shall be in accordance with MIL-STD-2073.

5.1.1 Level C. Preservation, Packaging and Packing shall be in accordance with MIL-STD-1190.

5.2 Marking. Each container of primer shall be legibly and durably marked in accordance with MIL-STD-129. Marking information shall include, but shall not necessarily be limited to, the following:

a. Title, number and date of this specification

b. Lot or batch number

c. Mixing ratios

d. Necessary precautionary notices.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The primer covered by this specification is intended for use with an adhesive tape which will be pressed between two metal surfaces primed as specified herein.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

c. When first article is required (see 3.1 and 4.4)

d. Whether toxicity certification is required (see 4.7.9)

e. Levels of preservation, packing and marking (see 4.8 and section 5).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified in 4.3. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Metrication. Wherever inch/pound dimensions are used in this document, metric equivalents in accordance with FED-STD-376 are acceptable.

6.5 Subject term (keyword) listing.

Adhesive, undercoating
Basecoat, adhesion
Compound, basecoat
Resin, nitrile-phenolic
Sealant, undercoat

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-47276B (MI)	2. DOCUMENT DATE (YYMMDD) 910925

3. DOCUMENT TITLE
PRIMER, BONDING

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDER U.S. ARMY MISSILE COMMAND	b. TELEPHONE (Include Area Code) (1) Commercial: (205) 876-6980 (2) AUTOVON: 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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